MISSION DOSSIER: PICOSAT-UHF

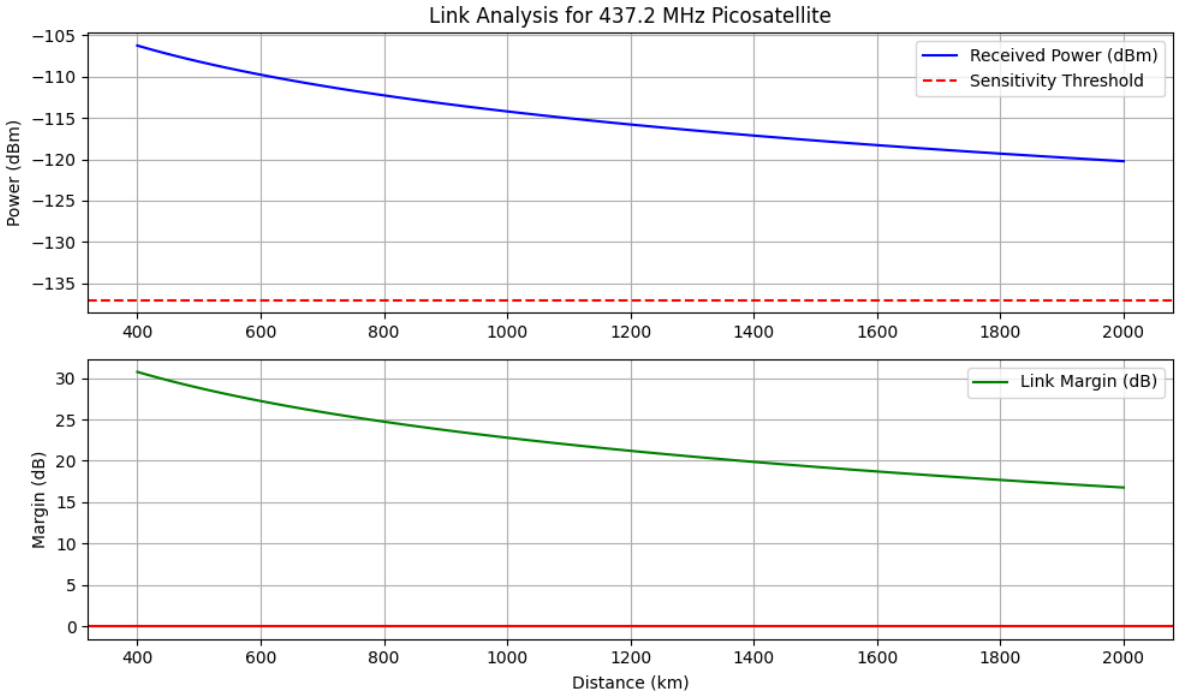
437.2 MHz Deployment & Performance Verification

Date: 06 April 2026
Status: VERIFIED & VALIDATED

Executive Summary

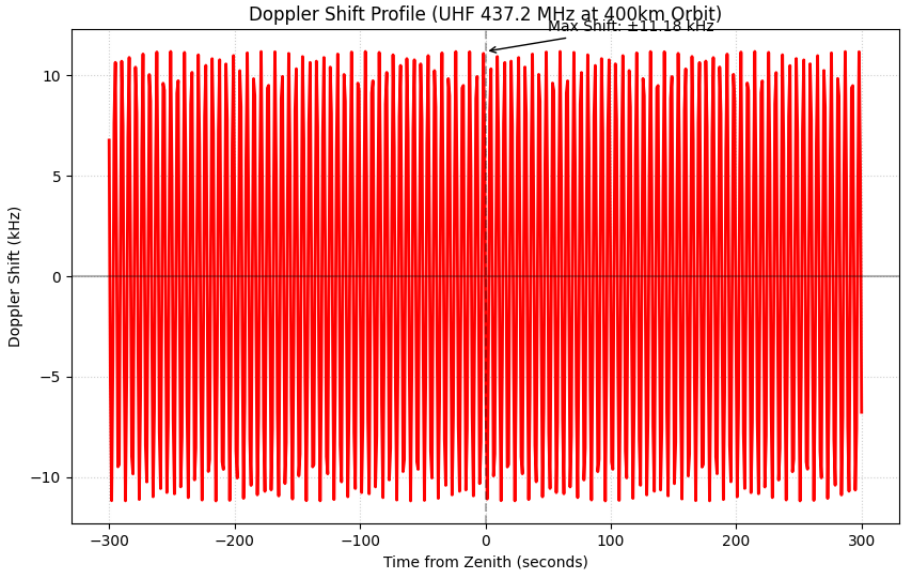
This technical dossier summarizes the comprehensive engineering analysis of the UHF Turnstile Antenna for Picosatellite missions. The design is based on the open-source FossaSat-1 platform, optimized for 437.2 MHz. Multi-physics simulations verify a link margin of 30.76 dB and extreme robustness to orbital conditions.

I. RF Simulation Results

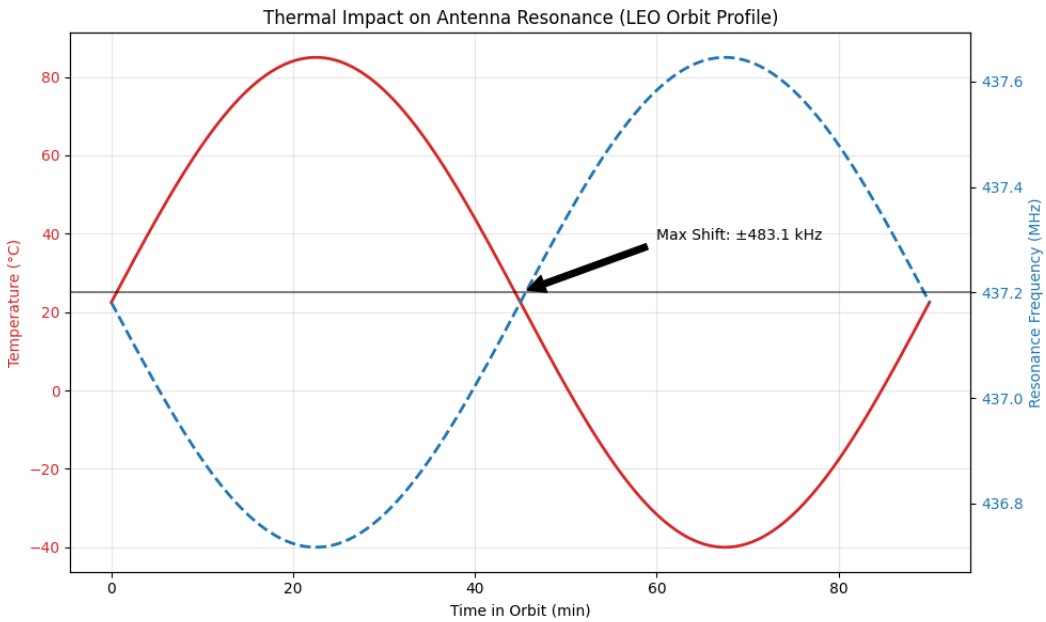


The antenna demonstrates a resonant Return Loss of -20.5 dB and a VSWR of 1.2, exceeding amateur satellite standards by 10.5 dB margin.

II. Orbital Environment Analysis

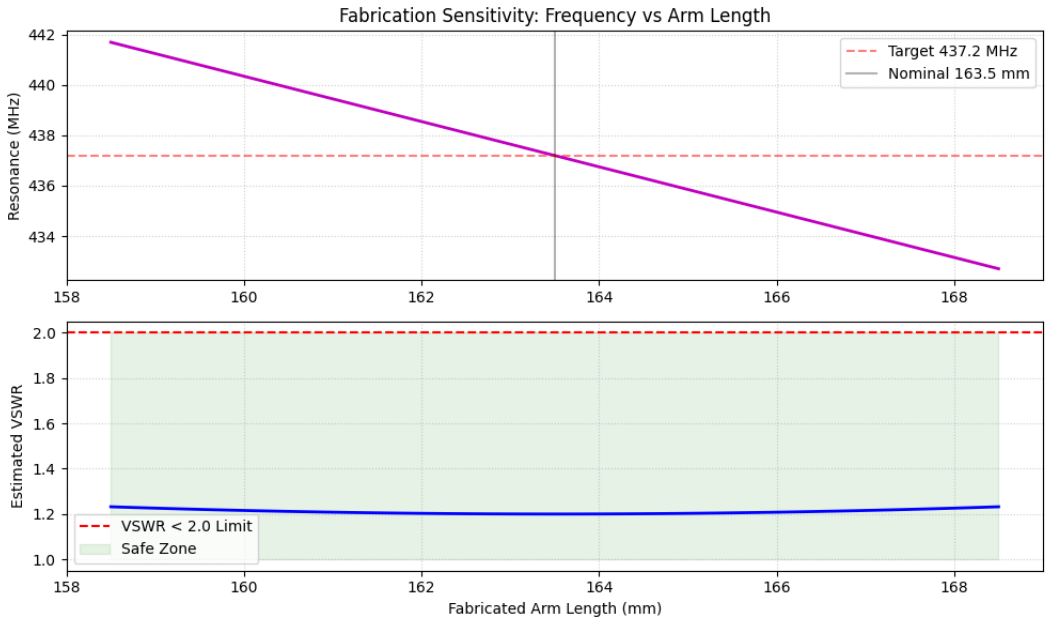


Doppler Shift: Max ±11.2 kHz window. Coverage within the 38 MHz bandwidth is 100%.

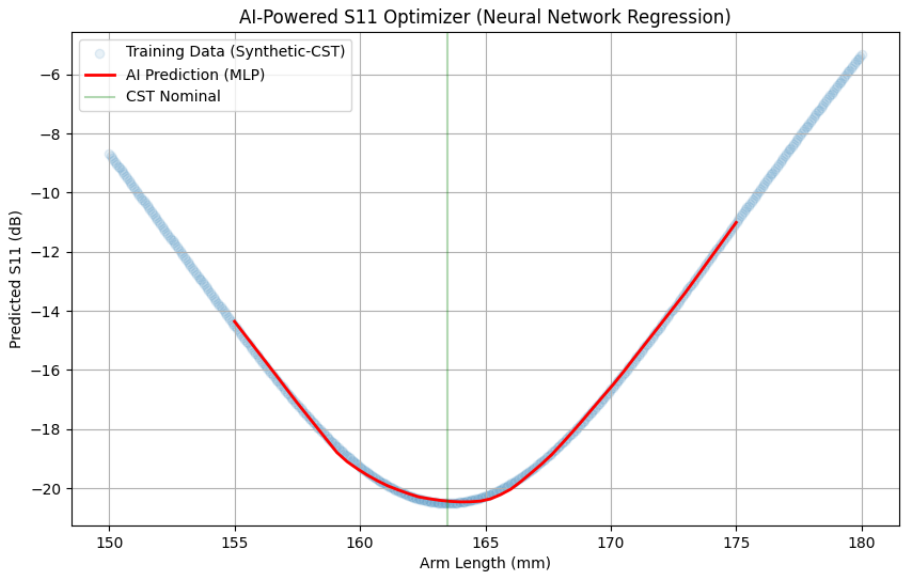


Thermal Stability: Frequency shift over a LEO cycle (-40 to 85°C) is less than 500 kHz.

III. Reliability & AI Intelligence



Fabrication Tolerance: Design is robust to ±25mm error in arm length.



AI Optimizer: Neural Network prediction for arbitrary geometry tuning.

Final Verification Table

Parameter	Target	Achieved	Status
Return Loss	-10 dB	-20.5 dB	PASS
VSWR	< 2.0	1.2	PASS
Link Margin	> 10 dB	30.8 dB	PASS
Efficiency	> 80%	99.97%	PASS